

Developmental Engineering Project

Functional Regime Detection in Limit Order Book Data

LOB Dataset Only

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DEP Endsem Evaluation

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1. Motivation: Why This LOB Project Matters

- Limit order books reveal the nearest observable state of market supply and demand because they contain standing liquidity, queue asymmetry, and price-level-specific depth information before execution occurs.
- Price-only systems usually react after the microstructure has already changed; by contrast, order-book geometry often reflects instability earlier through thinning liquidity, skewed replenishment, and directional pressure.
- The developmental engineering goal is to turn this high-frequency market state into a measurable early-warning system that is mathematically interpretable and practically usable in monitoring interfaces.
- The project is therefore not only about prediction. It is about **detecting structural transitions** in market organization using principled statistical representations rather than ad hoc heuristics.

2. Problem Statement, Objective, and Dataset Scope

Objective

Construct an LOB-only pipeline that transforms ten-level order-book snapshots into functional objects, embeds them in a compact latent space, and detects regime shifts using rigorous multivariate statistics.

- The source file contains 149 columns: 40 LOB columns, 40 technical-indicator columns, 64 time-series-derived columns, and 5 labels.
- This pipeline intentionally focuses only on the LOB-specific quantities: ten bid volumes, ten ask volumes, ten bid prices, and ten ask prices.
- The cached repository state contains 39,512 snapshots, each snapshot representing a ten-level book state.

3. Mathematical Representation of One Snapshot

For snapshot k , the book is modeled as depth-indexed functions over $i \in \{1, \dots, 10\}$:

$$q_k^b(i), \quad q_k^a(i), \quad p_k^b(i), \quad p_k^a(i)$$

where q denotes queued volume and p denotes quoted price.

- This converts the feature space from an unstructured 40-dimensional row into a structured object over a depth domain.
- Nearby levels are economically linked, so treating the book as a function across level is more faithful than treating each level as unrelated.
- The price arrays support spread and diagnostic metrics, while the volume arrays drive the main functional channels used in detection.

4. Functionalization Into Five Channels

For each snapshot k , the pipeline constructs:

$$\begin{aligned}f_k^b(i) &= q_k^b(i), & f_k^a(i) &= q_k^a(i) \\g_k^b(i) &= \sum_{j \leq i} f_k^b(j), & g_k^a(i) &= \sum_{j \leq i} f_k^a(j) \\h_k(i) &= \frac{f_k^b(i) - f_k^a(i)}{|f_k^b(i)| + |f_k^a(i)| + \varepsilon}\end{aligned}$$

- The first two channels store local bid and ask depth.
- The cumulative channels summarize the total liquidity profile as one moves deeper into the book.
- The imbalance channel normalizes asymmetry and keeps the scale bounded, which improves comparability across snapshots.
- The output per snapshot is a 5×10 tensor.

Website Image Slide A: LOB Overview or Snapshot Inspector

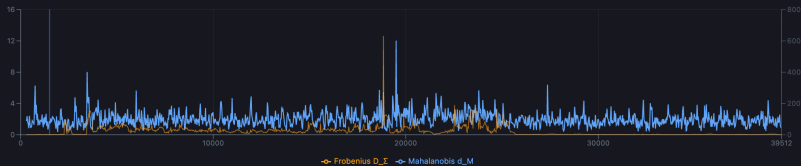
Limit Order Book Monitor

5-channel functional representation of 10-level depth · per-channel B-spline + PCA · Hotelling T^2 change-point detection

Ref 300 Test 30 Min d_M 3 Min consec 2 Snapshot 1500

Regime Timeline — Hotelling T^2 & Frobenius D_{Σ}

39,512 snapshots · 1307 windows · 31 bands · $\alpha_{adj}=7.7e-6$



Online Alarm — Mahalanobis² vs rolling reference



Red: per-snapshot Mahalanobis² in latent space (rolling ref=300). Blue: CUSUM of excursions above $2\chi^2_{0.99}$.

5. Why Functional Data Analysis Is Appropriate

- The order book is naturally ordered by level, and adjacent levels often vary smoothly rather than independently.
- FDA respects this structure by representing the snapshot as a curve-like object over depth.
- This reduces sensitivity to raw level noise while preserving larger geometric properties such as slope, convexity, and asymmetry.
- It also creates a clean bridge to downstream statistical inference: raw curves become basis coefficients, and basis coefficients become latent scores that can be compared with classical multivariate tests.

6. B-Spline Basis Expansion

Each of the five channels is observed on the grid $\{1, \dots, 10\}$ and approximated with a B-spline basis:

$$x(i) \approx \sum_{r=1}^q c_r B_r(i)$$

where $q = 4$ by default.

- The coefficients $c_1, \dots, c_q$ summarize each channel with a low-dimensional smooth representation.
- Separate spline fits are used for each channel so bid shape, ask shape, cumulative depth, and imbalance are not blended.
- With 5 channels and 4 basis functions per channel, one snapshot is compressed into a 20-dimensional coefficient vector.

7. PCA Embedding of the Functional Coefficients

Let $c_k \in \mathbb{R}^{20}$ be the concatenated coefficient vector for snapshot k . PCA computes orthogonal directions $v_1, \dots, v_m$ and maps

$$z_k = ((c_k - \bar{c})^\top v_1, \dots, (c_k - \bar{c})^\top v_m)$$

with $m = 5$ by default.

- The first reference block is used to fit the PCA basis, which avoids information leakage from future states.
- This latent vector is the main state representation used for change detection.
- In the cached project output, the first five components explain most of the variance, with the first component carrying roughly 35.8%, and the 5 Principle Components carrying about 90%.

8. Reference Window and Test Window Methodology

The detector scans latent scores using:

$$L = 300 \quad \text{reference points}, \quad M = 30 \quad \text{test points}$$

For scan position t :

$$Z_r = \{z_{t-L}, \dots, z_{t-1}\}, \quad Z_t = \{z_t, \dots, z_{t+M-1}\}$$

- The reference block estimates the recent baseline regime.
- The test block estimates the next candidate regime.
- Sliding by the test-window length creates a sequence of local regime-comparison experiments across the dataset.

9. Hotelling T^2 for Latent Mean Shift

For latent means μ_r, μ_t and pooled covariance Σ_p :

$$\Sigma_p = \frac{(L-1)\Sigma_r + (M-1)\Sigma_t}{L+M-2}$$

$$T^2 = \frac{LM}{L+M}(\mu_t - \mu_r)^\top \Sigma_p^{-1}(\mu_t - \mu_r)$$

- This is the multivariate analogue of comparing two means while accounting for covariance structure.
- Large T^2 indicates that the test window occupies a different latent center than the reference window.
- The covariance-adjusted geometry matters because movement along a stable direction should count more than movement along a high-variance direction.

10. fANOVA and Covariance Drift Statistics

The detector adds two more views of structural change.

$$F = \frac{SSB/df_b}{SSW/df_w}$$

where SSB is the between-window sum of squares and SSW is the within-window sum of squares.

$$D_{\Sigma} = \|\Sigma_t - \Sigma_r\|_F$$

- The F -statistic asks whether the two latent clouds separate more than their internal dispersion would normally allow.
- The Frobenius norm D_{Σ} measures how the covariance geometry itself changes, even when the means alone do not fully describe the transition.
- Taken together, these statistics detect both center shift and structural reshaping.

11. Decision Rule: Significance, Effect Size, and Persistence

A window is declared a structural shift only when multiple conditions are met:

$$p_{T^2} < \alpha_{\text{adj}}, \quad p_F < \alpha_{\text{adj}}, \quad d_M \geq 3.0$$

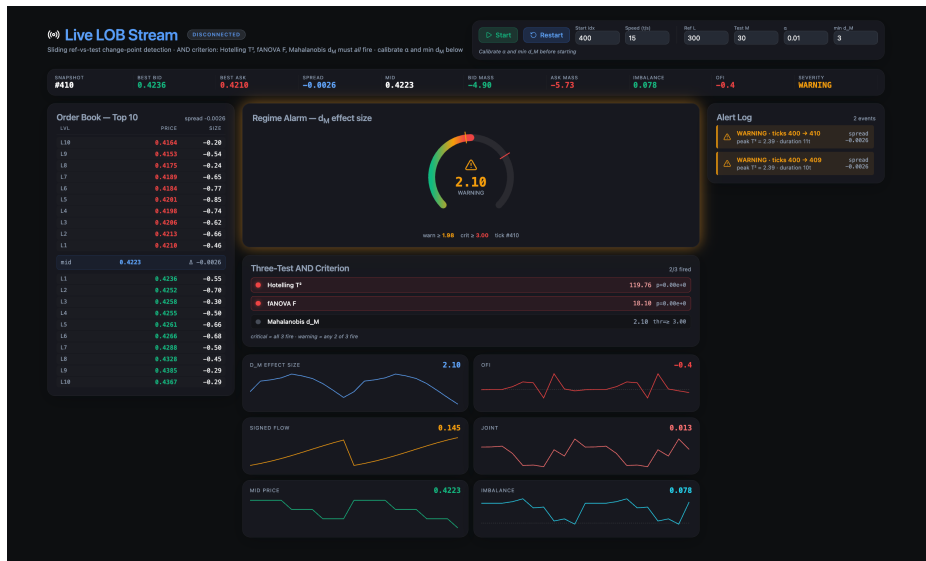
where

$$d_M = \sqrt{T^2 \frac{L + M}{LM}}$$

is a normalized Mahalanobis effect size.

- Bonferroni correction is used so $\alpha_{\text{adj}} = \alpha / n_{\text{windows}}$. There are 1307 scan windows, so there are 1307 $((39512-300)/30 = 1307)$ tests in the Bonferroni family.
- A persistence filter then keeps a flag only if it belongs to a run of at least two consecutive flagged windows.
- This sharply reduces spurious alerts from isolated noise bursts.

Website Image Slide B: Regime Timeline or Alarm Panel



12. Methodology Summary and Tech Stack

- 1 Load ten-level bid/ask prices and volumes from the LOB dataset.
- 2 Transform each snapshot into five functional channels.
- 3 Fit per-channel B-spline bases and concatenate the coefficients.
- 4 Fit PCA on a reference segment and project every snapshot into latent space.
- 5 Compare reference and test windows using Hotelling T^2 , fANOVA, and covariance drift.
- 6 Apply Bonferroni correction, effect-size filtering, and persistence filtering.
- 7 Produce regime bands, online alarms, and label diagnostics.
- 8 Expose these outputs through backend endpoints and dashboard components.

Core stack:

- **scikit-fda**: functional data objects and B-spline basis conversion.
- **scikit-learn**: PCA for latent embedding.
- **SciPy**: probability distributions and significance computations.
- **FastAPI**: APIs for LOB scan, snapshot view, window diagnostics, latent trajectories, microstructure summaries, imbalance grids, order-flow signals, and live streaming.
- **React frontend**: pages for regime overview, snapshot inspection, statistics panels, charts, and comparison tools.

Website Image Slide C: Dashboard Functionality Overview

LOB Snapshot Inspector

Drill a single ref-vs-test window — inspect pre/post depth, label drift (KL), latent mean shift

End idx 2000 Ref 300 Test 30 **Analyze**

Depth Profile — mirrored book + smooth B-spline



Bid mass	Ask mass	Imbalance
-5.29	-6.11	0.081
Center thick	Spread	Conv bid
-1.13	-0.0096	0.88

Depth Profile — mirrored book + smooth B-spline



Bid mass	Ask mass	Imbalance
-5.53	-6.21	0.058
Center thick	Spread	Conv bid
-1.13	0.0025	2.19

Pre / Post Depth — ref vs test mean



$\Delta \text{bid} / \Delta \text{ask}$ per level: L1: -0.22 · L2: -0.19 · L3: -0.14 · L4: -0.05 · L5: -0.12 · L6: -0.10 · L7: +0.01 · L8: -0.09 · L9: -0.06 · L10: +0.24

Regime Verdict

ref [1671, 1970] · test [1971, 2000]

STABLE

Hottelling T²

156.427

p<1e-6

Frobenius O.E

1.912

covariance

RANOVA F

26.343

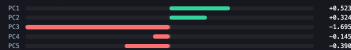
p<1e-6

n ref / test

300 / 30

snapshots

LATENT MEAN SHIFT



13. Practical Use Cases

- **Execution monitoring:** detect when visible liquidity becomes unstable before routing larger orders.
- **Intraday surveillance:** identify persistent shifts in book structure that may reflect news, stress, or adverse selection.
- **Adaptive strategy switching:** change quoting aggressiveness, participation, or inventory logic when the latent regime departs from the recent baseline.
- **Market-quality analysis:** study how liquidity geometry changes even when immediate price movements are modest.
- **Research and validation:** compare regime events with future-label distribution changes to evaluate whether structural shifts have predictive consequences.

14. Conclusion

Takeaway

This project treats the limit order book as a functional state space, compresses that state space into a stable latent representation, and detects market-structure transitions with statistically interpretable regime tests.

- The mathematical contribution is the bridge from microstructure geometry to functional representation and multivariate change detection.
- The engineering contribution is the translation of that pipeline into cached computations, APIs, live alarms, and interactive visual inspection.
- The practical contribution is earlier and more structured awareness of liquidity regime change than a price-only monitoring system can provide.